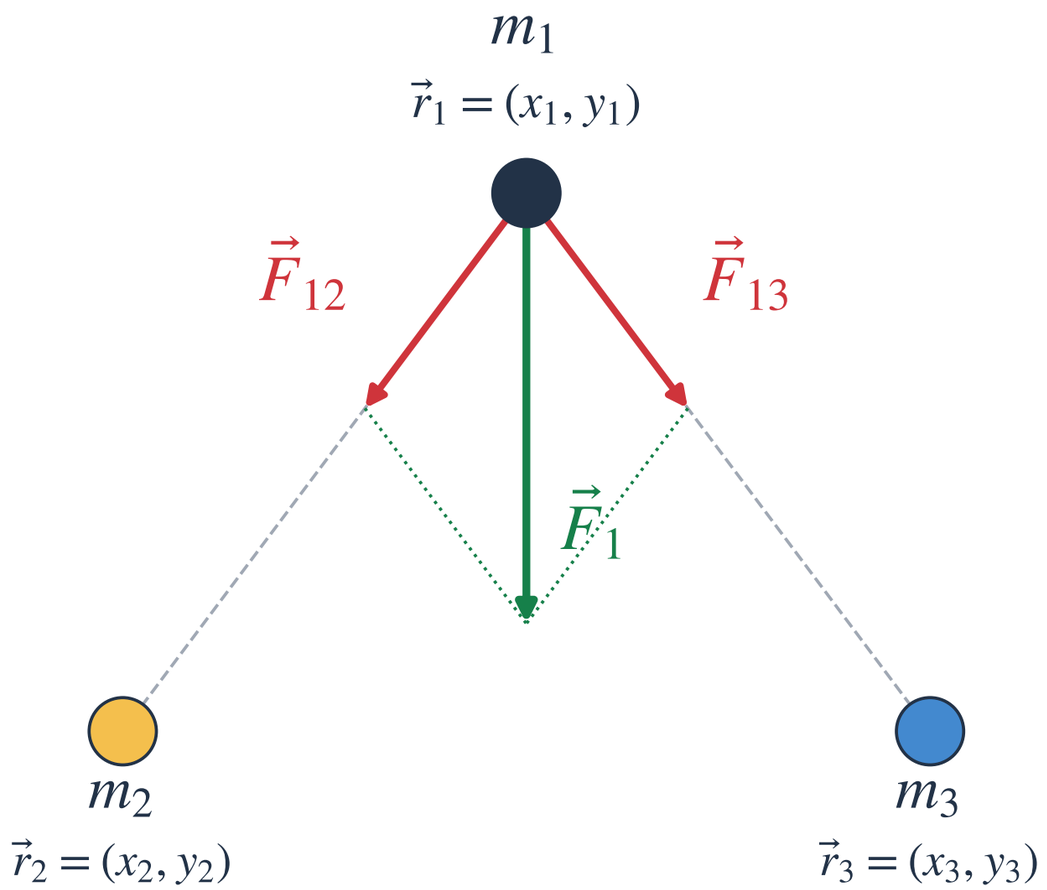
Gravitational forces: adding the contributions



$$\vec{F}_{12} = -\frac{Gm_1m_2}{(\Delta r_{12})^3}(\vec{r}_1 - \vec{r}_2)$$

Force on body 1 from body 2.

$$\vec{F}_{13} = -\frac{Gm_1m_3}{(\Delta r_{13})^3}(\vec{r}_1 - \vec{r}_3)$$

Force on body 1 from body 3.

$$\vec{F}_1 = \vec{F}_{12} + \vec{F}_{13}$$

$$\Delta r_{1j} = |\vec{r}_1 - \vec{r}_j|$$

All force arrows shown act on body 1; each pairwise force points toward the attracting body.

Illustration: equal masses and equal separations. The net force is the vector sum.